

Licheng Luo

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Google Scholar — Personal Website

Research Interests

Robotics, reinforcement learning, formal methods, spatio-temporal logic, robot learning, motion planning

Education

University of California, Riverside

Ph.D. in Mechanical Engineering, GPA: 4.0/4.0

Riverside, CA

Jan. 2025 – Present

University of Virginia

M.S. in Computer Science, GPA: 4.0/4.0

Charlottesville, VA

Aug. 2023 – Dec. 2024

Huazhong University of Science and Technology

B.Eng. in Electronic Information Engineering, GPA: 3.78/4.0

Wuhan, China

Sep. 2019 – Jun. 2023

Selected Publications

1. **L. Luo**, K. Liang, C. I. Vasile, and M. Cai. “Parallel Monitoring of Differentiable Spatio-Temporal Logic for Robot Manipulation.” Manuscript on arXiv, 2026.
2. **L. Luo**, K. Liang, C. I. Vasile, and M. Cai. “Differentiable SpaTiaL: Symbolic Learning and Reasoning with Geometric Temporal Logic for Manipulation Tasks.” Submitted to *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026.
3. **L. Luo**, Y. Xia, K. Liang, and M. Cai. “NL2SpaTiaL: Generating Geometric Spatio-Temporal Logic Specifications from Natural Language for Manipulation Tasks.” Submitted to *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026.
4. K. Liang, **L. Luo**, Y. Wang, M. Cai, and C. I. Vasile. “Time-aware Motion Planning in Dynamic Environments with Conformal Prediction.” *Learning for Dynamics and Control (L4DC)*, 2026. Oral presentation.
5. H. Wang, **L. Luo**, Y. Kantaros, B. Sinopoli, and M. Cai. “Deadlock-Free Hybrid RL-MAPF Framework for Zero-Shot Multi-Robot Navigation.” Manuscript on arXiv, 2025.
6. **L. Luo** and M. Cai. “Bridging Deep Reinforcement Learning and Motion Planning for Model-Free Navigation in Cluttered Environments.” Manuscript on arXiv, 2025.
7. **L. Luo**. “Policy Optimization in Robust Markov Decision Processes with Transition Gradient Theorem.” Master’s thesis, University of Virginia, 2024.
8. Y. Chen*, H. He*, **L. Luo***, K. Liu, M. Jiang, S. Li, X. Zhang, X. Yang, and Q. Liu. “Studying Pulmonary Fibrosis Due to Microbial Infection via Automated Microscopic Image Analysis.” *Frontiers in Microbiology*, 2023.

Research Experience

Research Assistant, University of California, Riverside

Jan. 2025 – Present

Advisor: Mingyu Cai

- Develop differentiable SpaTiaL methods for robot task specification, evaluation, and learning.
- Build tensorized geometric reasoning modules in PyTorch for differentiable spatio-temporal semantics.
- Study robustness signals for reinforcement learning, imitation learning, and policy evaluation.

Researcher, University of Virginia

Aug. 2023 – Dec. 2024

Advisor: Shangdong Zhang

- Studied robust Markov decision processes and transition-gradient-based optimization.
- Worked on motion planning, simulation, and learning-based control for robotic systems.

Undergraduate Researcher, MCLab, Huazhong University of Science and Technology

Wuhan, China

Advisor: Xin Yang

- Conducted undergraduate research in image analysis and machine learning applications.
- Contributed to microscopy-image-based analysis leading to a peer-reviewed publication.

Teaching

Teaching Assistant, University of California, Riverside

Riverside, CA

ME145, Fall 2025; ME018B, Spring 2026

Honors and Skills

Honors: Graduated with Honors, HUST, 2023; Merit-Based Academic Scholarship, 2019–2023

Programming: Python, C++, MATLAB, LaTeX, Bash

Tools: PyTorch, ROS, Unity, PyBullet, Isaac Gym, Stable-Baselines3, Docker, Linux, WandB